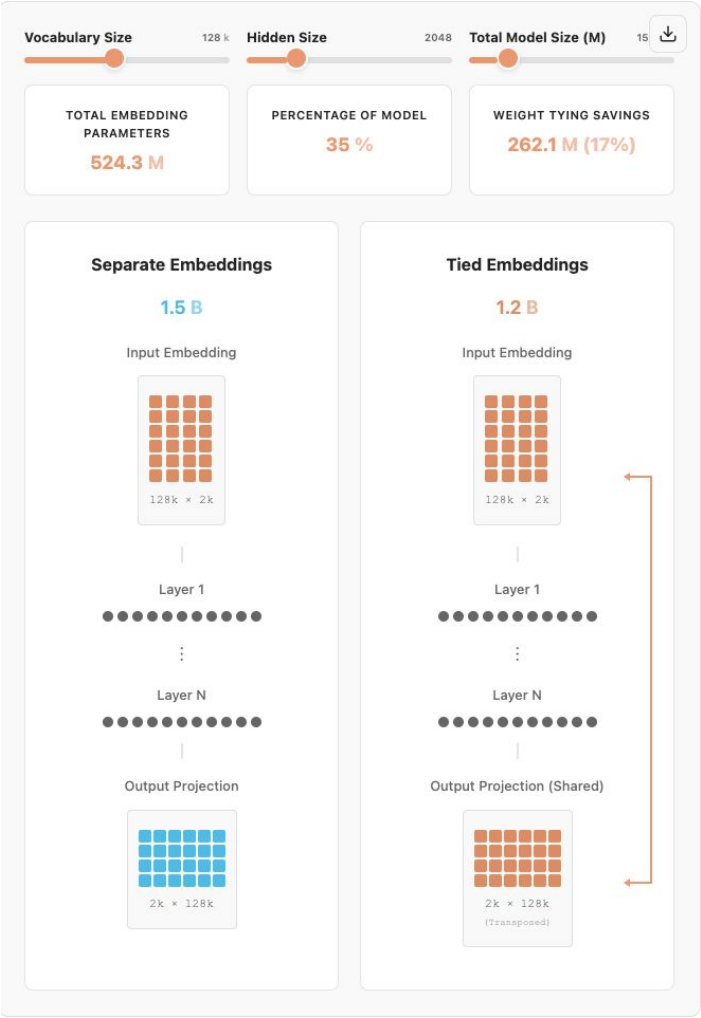
Embedding Sharing, Intro to Positional Encoding

The Smol Training Playbook


Vocabulary Size
128 k
Hidden Size
2048
Total Model Size (M)
15
TOTAL EMBEDDING PARAMETERS
524.3 M
PERCENTAGE OF MODEL
35 %
WEIGHT TYING SAVINGS
262.1 M (17%)
Separate Embeddings
1.5 B
Input Embedding
128k × 2k
Layer 1
Layer N
Output Projection
2k × 128k
Tied Embeddings
1.2 B
Input Embedding
128k × 2k
Layer 1
Layer N
Output Projection (Shared)
2k × 128k
(Transposed)

Choose a Model

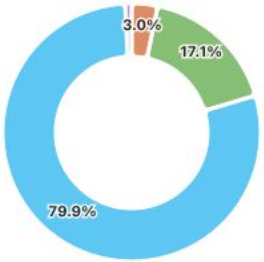
Llama 3.2 1B	SmolLM3 3B	Llama 3.1 8B	**Llama 3.1 70B**

Export JSON | View Original

70.55 B

Parameters

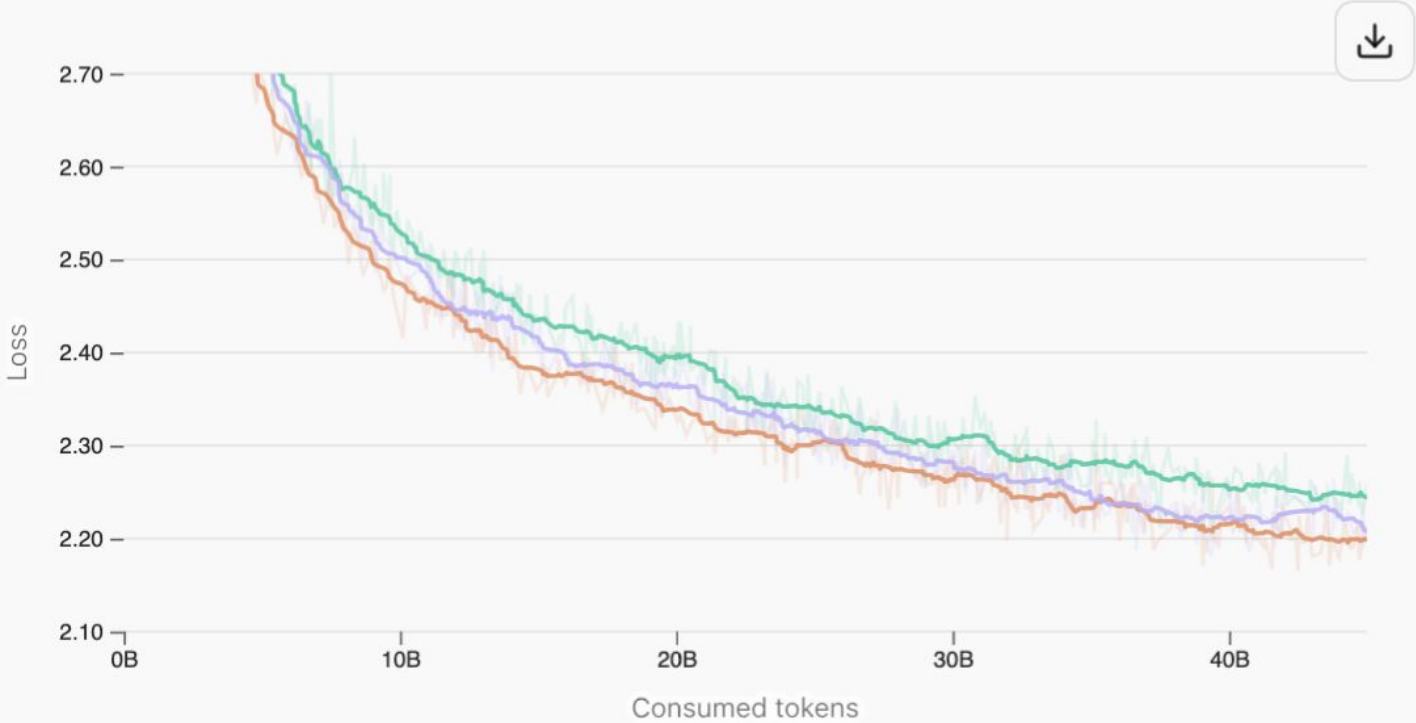
LAYERS	HIDDEN SIZE	HEADS (Q/KV)	INTERMEDIATE	EMBEDDINGS
80	8192	64/8	28672	Separate



Legend

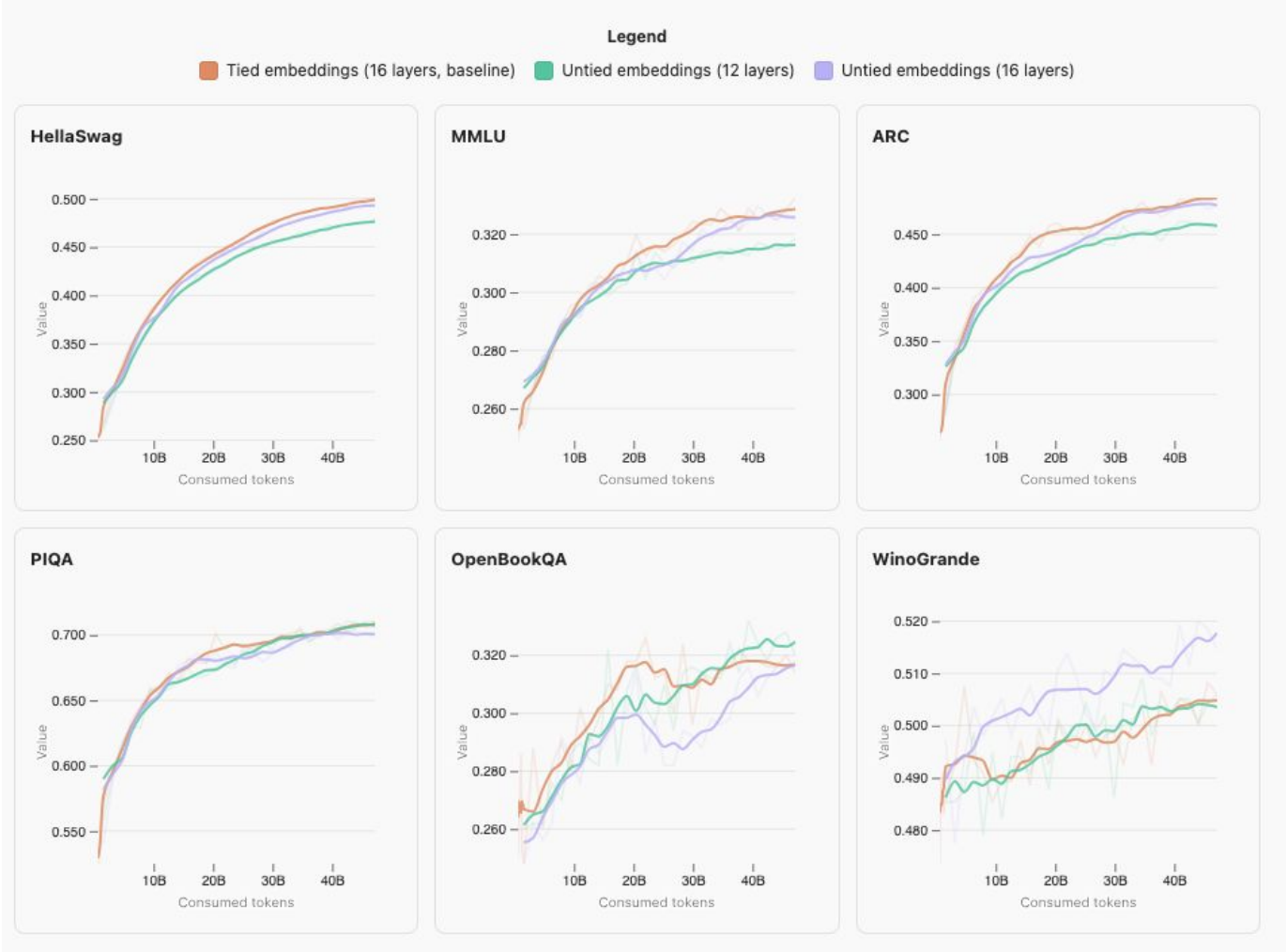
Embeddings | Attention | Feed Forward | Layer Norms

1.2B, 16 layers, tied = 1.2B, 12 layers, untied



Legend

- Tied embeddings (16 layers, baseline)
- Untied embeddings (12 layers)
- Untied embeddings (16 layers)


Legend
Tied embeddings (16 layers, baseline)
Untied embeddings (12 layers)
Untied embeddings (16 layers)
HellaSwag
0.500
0.450
0.400
0.350
0.300
0.250
Value
10B
20B
30B
40B
Consumed tokens
MMLU
0.320
0.300
0.280
0.260
Value
10B
20B
30B
40B
Consumed tokens
ARC
0.450
0.400
0.350
0.300
Value
10B
20B
30B
40B
Consumed tokens
PIQA
0.700
0.650
0.600
0.550
Value
10B
20B
30B
40B
Consumed tokens
OpenBookQA
0.320
0.300
0.280
0.260
Value
10B
20B
30B
40B
Consumed tokens
WinoGrande
0.520
0.510
0.500
0.490
0.480
Value
10B
20B
30B
40B
Consumed tokens

EARLY TRANSFORMERS: ABSOLUTE POSITIONAL EMBEDDINGS & LIMITATIONS

VISUALIZING THE MECHANISM AND CONSTRAINTS (e.g., Vaswani et al., 2023)

1. HOW IT WORKS: LEARNED LOOKUP TABLE

INPUT SEQUENCE

The (1) quick (2) brown (3) fox (4) jumps (5)

TOKEN EMBEDDINGS

The, quick, fox, brown

LEARNED LOOKUP TABLE (POSITIONAL EMBEDDINGS)

Pos 1	Pos 1 Vector
Pos 2	Pos 2 Vector
Pos 3	Pos 3 Vector
⋮	⋮
(e.g., 8)	Pos 8 Vector

TOKEN EMBEDDINGS

The + [Position Vectors] → + + +

Input Tokens | Position Vectors | Token + Position Embeddings

Learned vectors for specific positions (1, 2, 3...) are added to token embeddings.

Worked fine for short sequences.

2. MAJOR LIMITATION: LACK OF GENERALIZATION

TRAINING (e.g., Max Length 8)

✓ ALL POSITIONS HAVE TRAINED VECTORS.

✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓

1 2 3 4 5 6 7 8

→ VALID REPRESENTATIONS

MODELS' MAX INPUT SEQUENCE LENGTHS LIMITED TO TRAINED MAX. NO OUT-OF-THE-BOX GENERALIZATION TO LONGER SEQUENCES.

INFERENCE (Longer Sequence, e.g., Length 12)

✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ? ? ? ✗

1 2 3 4 5 6 7 8 9 10 11 12

✗ NO EMBEDDINGS FOR UNSEEN POSITIONS (Pos > Max Trained).

✗ MODEL CAN'T DETERMINE RELATIONSHIPS FOR UNTRAINED INDICES.

Note: Vaswani et al., 2023

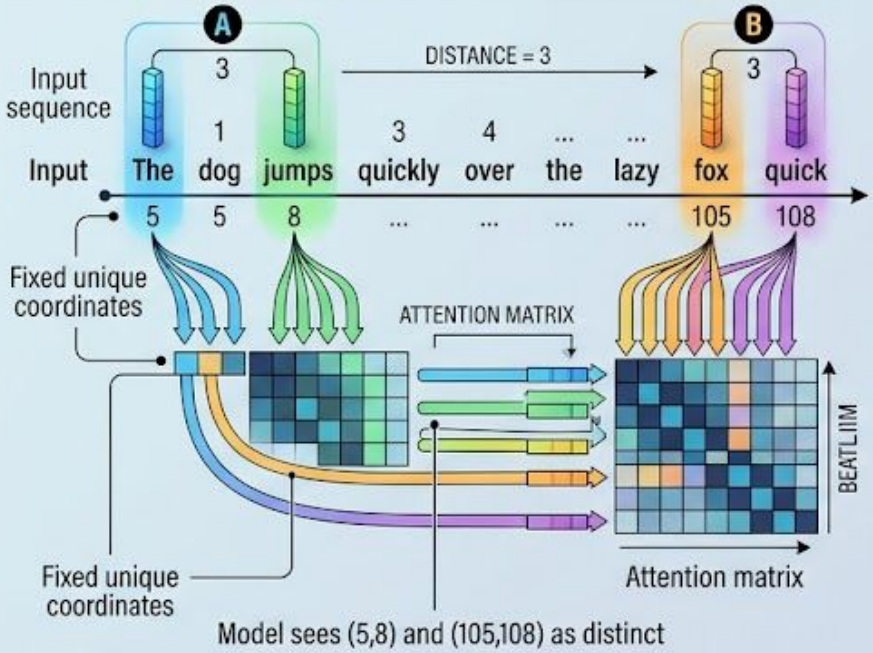
EVOLUTION OF TRANSFORMERS: ABSOLUTE TO RELATIVE POSITIONAL EMBEDDINGS

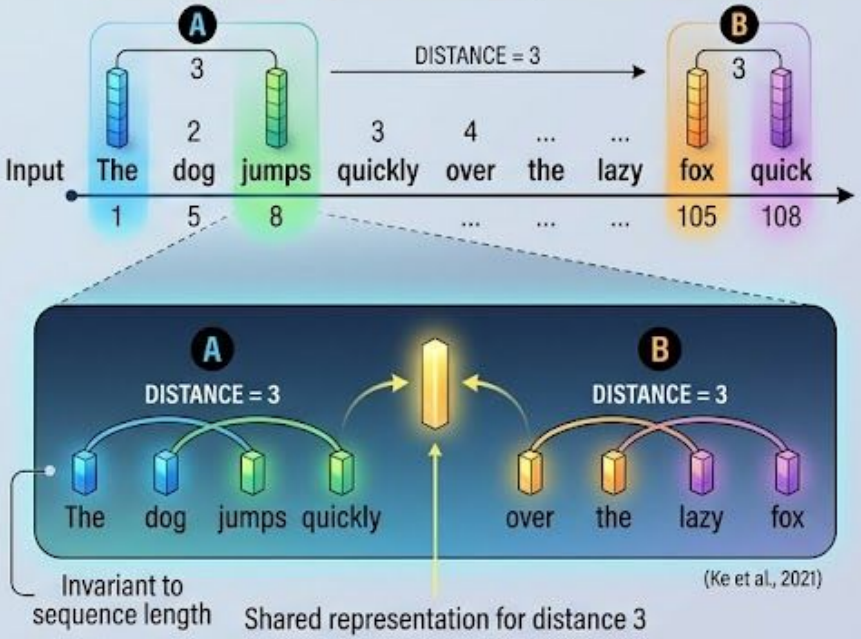
FOCUS ON DISTANCE BETWEEN TOKENS INSTEAD OF FIXED COORDINATES

1. ABSOLUTE POSITIONAL EMBEDDINGS
(OLDER METHOD)

EVOLUTION →

2. RELATIVE POSITIONAL EMBEDDINGS
(NEWER METHOD)

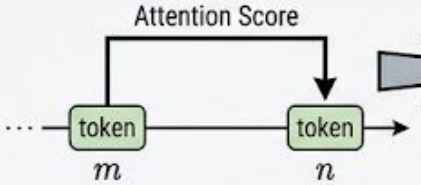




"Distance (3) isn't the primary feature."

"WHETHER TWO WORDS ARE 3 POSITIONS APART MATTERS MORE THAN (5,8) VS (105,108)."

THE GOAL: INJECT POSITION INFO INTO ATTENTION

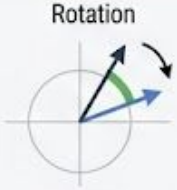


Attention Score depends only on the **relative offset** $m - n$, not absolute positions.

THE TRICK: ROTATION PRESERVES DOT PRODUCTS PREDICTABLY

$$\langle \mathbf{q}, \mathbf{k} \rangle \xrightarrow{\text{ROPE TRANSFORMATION}} \mathbf{R}(m\theta)\mathbf{q}$$

$$\langle \mathbf{q}, \mathbf{k} \rangle \xrightarrow{\text{TRANSFORMATION}} \mathbf{R}(n\theta)\mathbf{k}$$



$$\langle \mathbf{R}(m\theta)\mathbf{q}, \mathbf{R}(n\theta)\mathbf{k} \rangle = \mathbf{q} \cdot \mathbf{R}(n-m)\theta) \cdot \mathbf{k}$$

The inner product depends **ONLY** on the difference (n–m)

WHY IT WON (ADVANTAGES)

- No learned parameters
- No O(L²) bias matrix
- Automatic soft decay with distance
- Applied at every layer
- Extends to long contexts via Position Interpolation, YaRN

TL;DR

- Rotary Positional E Embedding (RoPE) useps
- No O(L²) bias matrix
- Automatic soft decay distance, at every layer
- Extends to long contexts via Position Interpolation, YaRN

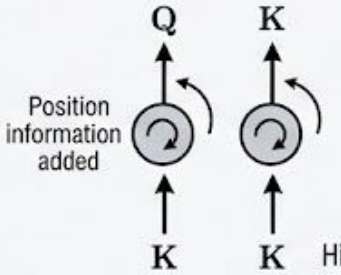
THE CONSTRUCTION: MECHANICS OF ROPE

1. **SPLIT** Vector (dim *d*) into *d*/2 Pairs

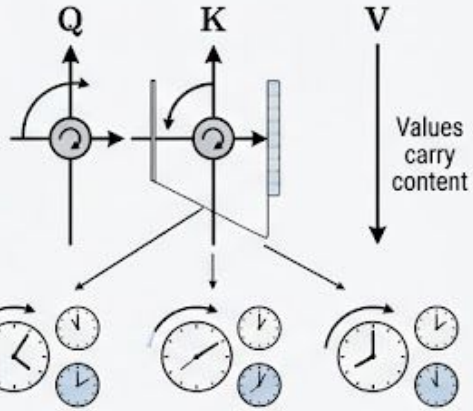
2. **ROTATE** pair *i* by Angle $m \cdot \theta_i$

$$\theta_i = 10000^{-2i/d}$$

3. **APPLY Rotation** to Q and K **ONLY**



3. **APPLY Rotation** to Q and **K ONLY**



Think of clocks ticking at different rates — the pattern of hands tells you the position.

Low-index pairs spin **FAST** (Fine local structure),
High-index pairs spin **SLOWLY** (Long-range structure)

IMPLEMENTATION

$$\mathrm{RoPE}(\mathbf{x}, m) = \underline{\mathbf{x} \odot \cos(m\theta)} + \underline{\mathrm{rotate_half}(\mathbf{x}) \odot \sin(m\theta)}$$

- Two **elementwise multiplies**
- Essentially free: cos and sin tables are precomputed

Define

$$f(x, m) = R(m\theta)\, x, \text{ where } R(\varphi) = [[\cos\varphi, -\sin\varphi],[\sin\varphi, \cos\varphi]].$$

Then

$$\langle R(m\theta)q, R(n\theta)k \rangle = q^\top R(m\theta)^\top R(n\theta)\, k = q^\top R((n-m)\theta)\, k,$$